Program - 9

Data transformation and Aggregation

Date : 10.10.2026

Aim

Write a Python program to perform data transformation and aggregation using Pandas, including grouping, feature engineering, and summarizing data

Algorithm

1. Start
2. Import the pandas library.
3. Create a dictionary containing student details:
   - Name
   - Department
   - Marks
   - Attendance
4. Convert the dictionary into a Pandas DataFrame named df.
5. Display the original student data.
6. Define a function assign_grade() to assign grades based on Marks:
   - Marks = 90 ? A+
   - Marks = 80 ? A
   - Marks = 70 ? B
   - Marks = 60 ? C
   - Marks < 60 ? D
7. Apply the assign_grade() function to the Marks column.
8. Create a new Grade column.
9. Calculate the Performance Score using 70% Marks and 30% Attendance.
10. Add the calculated Performance Score as a new column.
11. Display the dataset after feature engineering.
12. Group the students by Department.
13. Calculate department-wise:
    - Average Marks
    - Maximum Marks
    - Minimum Marks
    - Average Attendance
    - Average Performance Score
    - Number of Students
14. Display the department-wise summary.
15. Sort the students according to Marks in descending order.
16. Display the students sorted by Marks.
17. Calculate overall summary statistics for:
    - Marks
    - Attendance

- Performance Score

18. Display the overall summary statistics.
19. Stop.

##Program - 9

```
import pandas as pd
data = {
    'Name': ['Arun', 'Bala', 'Cathy', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'CSE', 'ECE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 72, 90, 65, 78, 88],
    'Attendance': [95, 80, 92, 75, 85, 90]
}
df = pd.DataFrame(data)
print("Original Data:")
print(df)
def assign_grade(marks):
    if marks >= 90:
        return 'A+'
    elif marks >= 80:
        return 'A'
    elif marks >= 70:
        return 'B'
    elif marks >= 60:
        return 'C'
    else:
        return 'D'
df['Grade'] = df['Marks'].apply(assign_grade)
df['Performance_Score'] = (
    df['Marks'] * 0.7 +
    df['Attendance'] * 0.3
)
print("\nAfter Feature Engineering:")
print(df)
department_summary = df.groupby('Department').agg(
    Average_Marks=('Marks', 'mean'),
    Maximum_Marks=('Marks', 'max'),
    Minimum_Marks=('Marks', 'min'),
    Average_Attendance=('Attendance', 'mean'),
    Average_Performance=('Performance_Score', 'mean'),
    Number_of_Students=('Name', 'count')
)
print("\nDepartment-wise Summary:")
print(department_summary)
sorted_df = df.sort_values(by='Marks', ascending=False)
print("\nStudents Sorted by Marks:")
print(sorted_df)
print("\nOverall Summary Statistics:")
print(df[['Marks', 'Attendance', 'Performance_Score']].describe())
```